

What open source blockchain platforms will help you learn

Open source blockchain platforms provide a transparent and collaborative environment for building decentralised applications (dApps). You can learn a lot by using them.

- **Understanding blockchain fundamentals:** You'll solidify your grasp of core blockchain concepts like consensus mechanisms (e.g., Proof of Work, Proof of Stake), cryptography, immutability, and distributed ledger technology.
- **Technical proficiency:** Depending on the platform, you'll develop skills in specific programming languages (e.g., Solidity for Ethereum), smart contract development, and working with blockchain developer tools.
- **Decentralised application development:** You'll learn the intricacies of designing and deploying dApps that leverage the trustless and transparent nature of blockchain.
- **Open source community engagement:** Open source platforms often have active communities. You'll learn to collaborate with developers, access resources, and contribute to the platform's growth.
- **Security considerations:** Blockchain security is paramount. You'll gain insights into potential vulnerabilities (e.g., smart contract exploits) and best practices for securing your dApps.
- **Scalability and performance:** Open source platforms can have varying levels of scalability and transaction processing speeds. You'll learn to assess these factors when choosing a platform for your project.
- **Interoperability:** Understanding how different blockchains interact and the challenges of integrating them is a valuable lesson.
- **Regulatory landscape:** The legal and regulatory environment surrounding blockchain is constantly evolving.

You'll gain awareness of relevant regulations that might impact your dApp.

By working with open source blockchain platforms, you'll not only build technical expertise but also gain a deeper understanding of the potential and limitations of this transformative technology.

What you will learn from private blockchain platforms

While open source blockchains offer a community-driven approach, private blockchains have their own set of learnings.

- **Focus on permissioned access:** You'll understand how to configure and manage user access within a private network. This includes defining user roles and permissions for data visibility and transaction participation.
- **Centralised governance:** Private blockchains often have a governing body. You'll learn how to navigate consortium dynamics and decision-making processes within that structure.
- **Customisation and integration:** Unlike open source platforms, private blockchains can be tailored to specific needs. You'll gain experience in customising the platform and integrating it with existing enterprise systems.
- **Performance optimisation:** Since scalability and speed are crucial in private environments, you'll learn optimisation techniques to ensure the platform meets your transaction volume and processing requirements.
- **Regulatory compliance:** Private blockchains are often used in regulated industries. You'll develop an understanding of how to configure the platform to comply with relevant data privacy and security regulations.
- **Vendor lock-in:** Some private blockchains are offered by specific

vendors. You'll learn to weigh the benefits of a controlled environment against potential vendor lock-in and limited interoperability.

- **Security in a trusted environment:** Security is still paramount, but the focus shifts from open network threats to access control and internal risk management within the trusted participants.

In summary, using private blockchains equips you with expertise in managing permissioned networks, navigating consortium governance, and optimising the platform for performance and compliance within a closed ecosystem.

Key learnings in using public blockchain platforms

Public blockchains, unlike their private counterparts, function in a completely open and permissionless environment. Here's what you'll learn by diving into this realm.

- **Open participation and transparency:** You'll gain first-hand experience in developing applications accessible to anyone with an internet connection. This fosters transparency and immutability of data on the public ledger.
- **Cryptoeconomics and tokenization:** Public blockchains often have their own cryptocurrencies or tokens used for network fees, rewarding participants, and even governing the platform. You'll develop an understanding of tokenomics — the design and economics behind these tokens.
- **Security in a distributed network:** Securing your dApps on a public blockchain requires a different approach. You'll learn about common vulnerabilities specific to public networks like phishing attacks or majority attacks, and how to mitigate them.

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